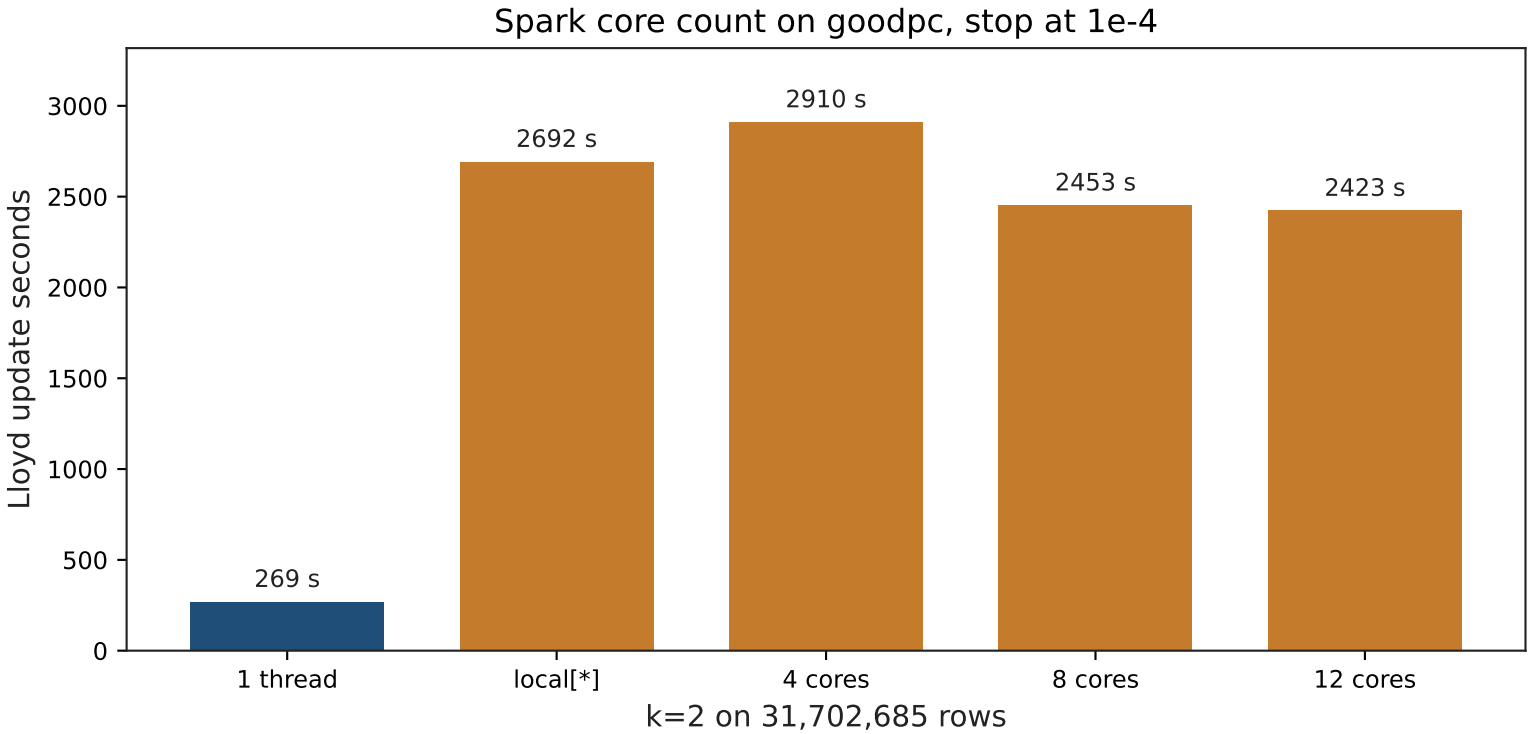

goodpc, AMD Ryzen 5 5600X, 6 cores / 12 threads, 31.89 GB. No separate VM.

local[*] is the earlier size-sweep run, 12 workers. It is not 1 core. 12 cores is the later local[12] run.

16-core Spark ran as local[12]. Wall clock starts after the matrix is loaded. Python is 1 worker. Same initial centres.

A finished CPU Spark local[1] file was not on disk.